VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Compliance Assessment Approach

AFLCMC/WA

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1.0 Overview

A unified and defined compliance approach allows developers and integrators to work independently using prescribed interfaces and touchpoints. This compliance approach increases the efficiency of overall system development. The compliance approach used by the Agile Mission Suite (AMS) Government Reference Architecture (GRA) allows for the use of legacy systems as well as newly developed systems, while minimizing rework and maintaining compatibility with the larger AMS GRA environment.

1.1 Purpose

The Compliance Assessment Approach Volume describes and defines the compliance approach for AMS GRA elements from individual Minimum Procurable Units (MPUs) to full Mission Systems. This volume also describes the use of external standards across the AMS GRA. Furthermore, the volume describes the roles and responsibilities MPU developers, Mission System Integrators (MSIs), and procuring authorities have in ensuring instantiations of AMS GRA are compliant. Finally, this volume includes the full set of compliance requirements for AMS GRA in **Chapter 5**.

1.2 Scope

The approach described in this Volume applies to any entity seeking AMS GRA compliance.

1.3 Authoritative References

Authoritative references are key documents that a procuring authority, MSI, or MPU provider refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. An AMS GRA entity is compliant when it provides the required prescriptive constructs (e.g. interfaces for an MPU). See **Table 1.3.1** for a list of authoritative references that guides compliance assessments.

Standard	Title	Version/Date
ISO/IEC/IEEE 29148:2018	Systems and Software Engineering — Life Cycle Processes — Requirements Engineering	2018-11-30
OMS Standard	OMS Description and Documentation	2.5

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing Compliance Assessments.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the Compliance Assessment Approach Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Executive Summary	14.0

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the Compliance Assessment Approach and Related Entities.

2.0 AMS GRA Compliance

The purpose of compliance is to achieve the architectural and business goals of AMS GRA while designing, developing, and integrating MPUs into a platform. These activities may involve merging existing and new capabilities, thereby driving a need to define multiple tiers of AMS GRA compliance. The AMS GRA approach includes two separate and distinct aspects: compliance tiers, and a weighted compliance score. The compliance tiers identify concrete levels that help ensure elements meet certain criteria to support integration goals. The separate score shows an element's overall progress toward full Tier 3 compliance. Using this approach, AMS GRA helps facilitate integration while simultaneously allowing MPUs to report a more granular level of adherence to the AMS GRA.

Compliance assessment occurs at various times during system development and is always associated with a specific version of the AMS GRA standard.

2.1 Relationship to OMS

AMS GRA leverages and builds on the architectural model and vision of the Open Mission Systems (OMS) standard in the following ways:

- Using OMS architectural elements such as the Abstract Service Bus (ASB), Critical Abstraction Layer, and OMS Service
- Building upon the OMS standard with required details to meet the needs of MSIs
- Adding to the OMS standard further by defining a common architecture and interfaces within the black box traditionally labeled an OMS Subsystem
- Adding concepts and compliance to meet the needs of the Mission Platform
- Adding the Avionics Backplane Bus (ABB) to support digital apertures
- Adding MPUs to account for independent procurement of additional architecture elements. MPU hardware and software interfaces are defined by the AMS GRA and implementations of MPUs must include non-proprietary documentation.
- Defining Development Kits (DKs) to support the different Processing Components (i.e. General Purpose Processor (GPP), Graphics Processing Unit (GPU), and Field-Programmable Gate Array (FPGA))

The OMS standard is independently managed and maintained, separate from AMS GRA. For an AMS GRA Mission System instantiation, the MSI, in coordination with the procuring authority, determines the specific version of OMS utilized. As such, OMS documentation is not distributed with AMS GRA.

A subset of the requirements defined by AMS GRA reference OMS interfaces, requirements, and templates. The OMS Service Contract is one such example. All OMS specific material can be found in the appropriate OMS distribution.

2.2 Compliance Approach

AMS GRA assesses two different kinds of compliance. The first type of assessed compliance is for MPUs. This assessment is necessary because MPUs are the architectural building blocks used to define and assemble an AMS GRA Mission System. The compliance of these MPUs with the architecture tenets directly affects the ability of the system architecture to achieve AMS GRA business and technical goals. Additionally, some MPUs require two assessments: one assessment against the MPU requirements and one against the MPU's Development Kit Requirements. See **Section 5.3** for more details on Development Kit Requirements.

The second type of assessed compliance is for the Mission System. This assesses key Mission System attributes that enable the AMS GRA system-level architecture and includes distinct requirements for Cyber and the Mission Agnostic Service Infrastructure (MASI).

Compliance verification tables include a Recommended Verification Method attribute for each compliance requirement. This method is chosen from among the verification methods defined in ISO/IEC/IEEE 29148:2018(E), Section 6.5.2.2, and summarized in the following excerpts. AMS GRA includes verification methods in the compliance tables; however, these methods are not prescriptive. An adopting program may alter the verification methods In Accordance With (IAW) the needs of a specific AMS GRA instantiation. For the purpose of compliance verification, the full definitions from IEEE 29148 apply as shown in **Figure 2.2.1**.

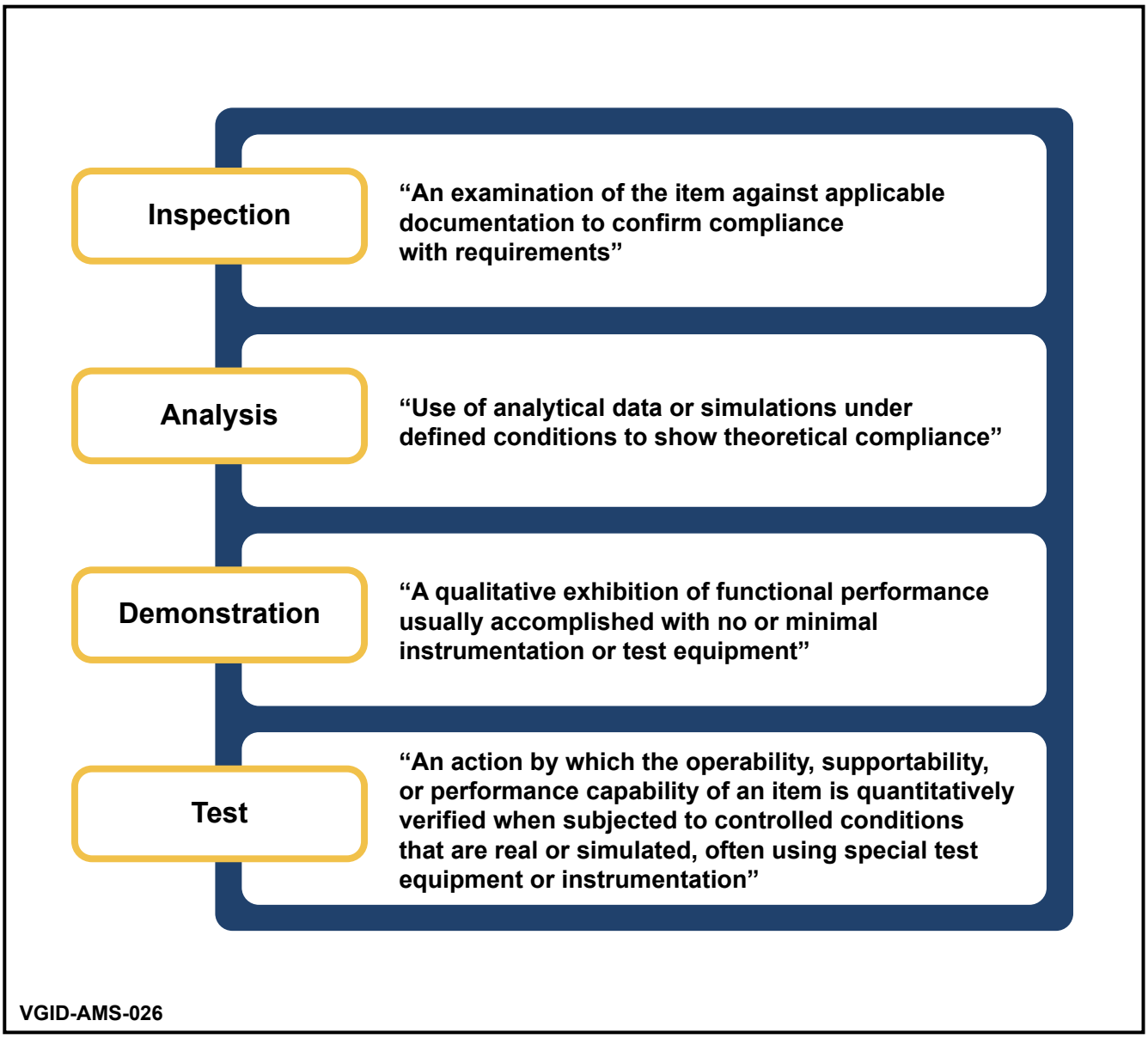


Figure 2.2.1: The Compliance Verification Definitions Graphic Describes the Inspection, Analysis, Demonstrate, and Test Verification Methods as Well as the Rigor of Analysis and Testing Required to Satisfy each Method.

Compliance assessment uses a combination of manual and automated methods to verify requirements for MPUs and Mission Systems in order to:

- Minimize time conducting compliance verification
- Reduce the recurring costs of compliance verification
- Increase product quality by removing human interaction and increasing assessment coverage

Compliance assessment mechanisms use the following approaches:

- **Inspection:** Use automation mechanisms and allow manual mechanism exceptions
- **Analysis:** Maximize automation mechanisms and allow manual mechanisms
- **Demonstration:** Use automation mechanisms and allow manual mechanism exceptions
- **Test:** Use automation mechanisms and allow manual mechanism exceptions

2.3 Basic Rules of Compliance

The AMS GRA compliance assessment follows the basic rules defined below for the two distinct aspects of compliance. As noted earlier, the two distinct aspects are compliance tiers and weighted scoring. By using both aspects, AMS GRA provides a straight-forward approach realizing benefits that would not be achievable by either approach on its own.

2.3.1 Compliance Tiers

Similarly to OMS, AMS GRA defines a set of compliance tiers that provide concrete steps toward full AMS GRA compliance. Meeting the requirements at a particular tier eases integration complexity and ensures alignment with specific AMS GRA goals. Each AMS GRA requirement is associated with a specific tier, and in order to be assessed at a specific tier, the element under assessment must meet all the applicable requirements at that tier and all lower tiers.

Figure 2.3.1 shows the considerations for distinction between each tier of an AMS GRA compliance requirement.

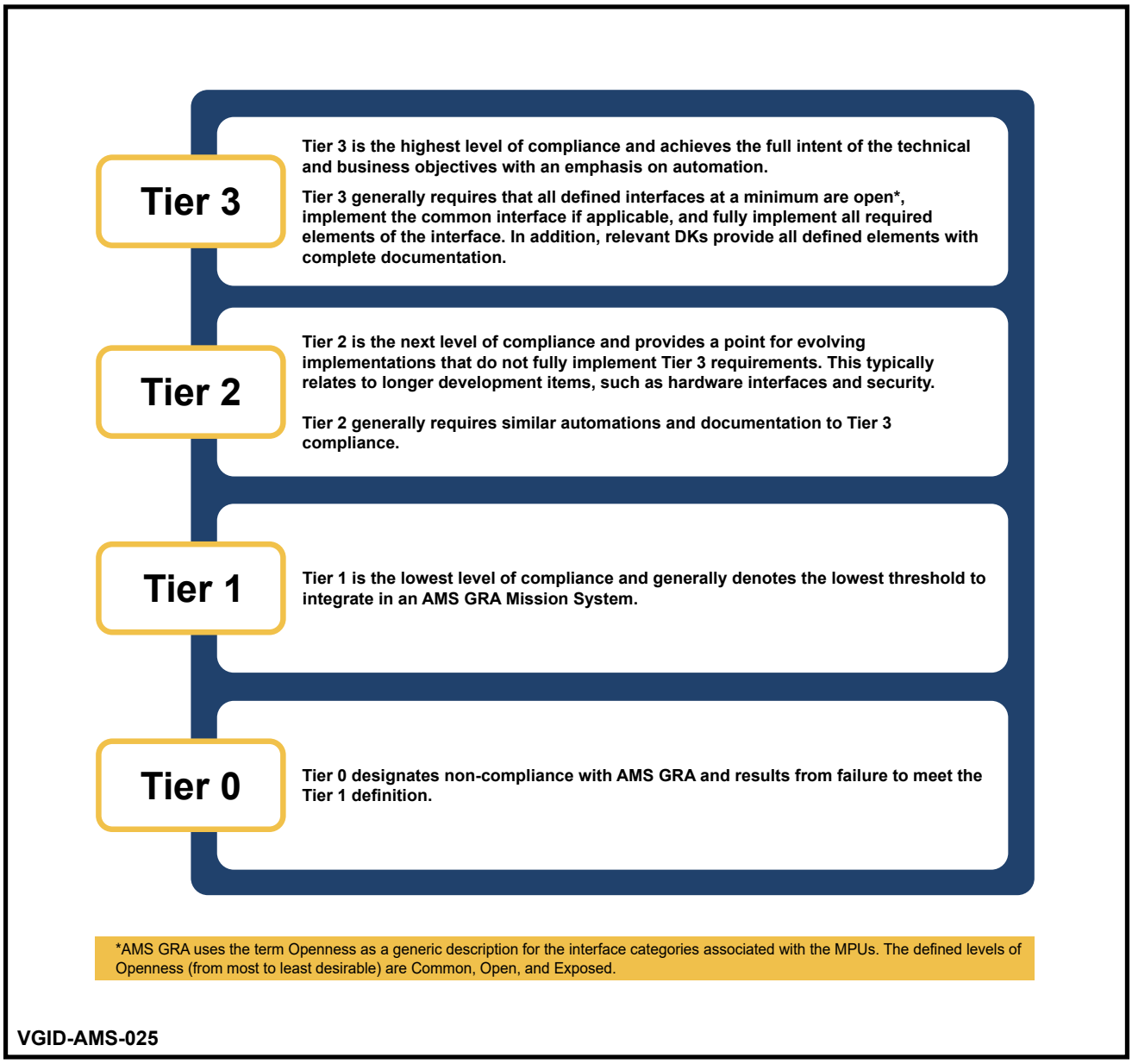


Figure 2.3.1: AMS GRA Compliance Requirement Tiering Defines the Levels of Compliance an MPU, Mission System, and DK Can Achieve.

2.3.2 Weighted Compliance Score

In addition to tiers, AMS GRA compliance uses the concept of a weighted compliance score. Each requirement has an associated weight which impacts how much that particular requirement affects the assessed score. This score is assessed from 0-100% and is independent from the tier assessment. Using a weighted score assessment provides several distinct benefits that are not realized from tiers alone:

- MPUs/Mission Systems can show incremental progress towards AMS GRA compliance even if a new tier is not achieved
- MPUs/Mission Systems receive credit for meeting requirements at higher tiers, even if they do not fully meet those tiers
- Procuring authorities have the ability to shape the scoring assessment to align with their system priorities (including tailoring out requirements that do not support the adopting program's needs)

Calculation of a weighted compliance score is simply the sum of the weights of the requirements satisfied by the element under assessment, divided by the sum of the weights of the requirements applicable to the element under assessment. The assessed compliance score for an MPU should increase primarily with improvements to the MPU that align with AMS GRA.

AMS GRA provides initial default weights which can be adjusted by procuring authorities. Because procuring authorities have the ability to modify the weights associated with requirements, a specific element under assessment may receive different scores from two different procuring authorities. This tailoring does not impact the tier assessment which is absolute based on the AMS GRA tier definitions.

2.3.3 Compliance Categories

As noted in **Section 2.2**, AMS GRA assesses compliance for both MPUs and Mission Systems.

For MPUs, applicable compliance requirements are captured in a table for each MPU. Compliance is documented for each applicable requirement resulting in both an assigned compliance tier, and a weighted compliance score. When applicable, compliance for Digital Payload Processor (DPP) and Multi-Function Aperture (MFA) MPUs is assessed for both the MPU and DK categories of requirements.

For Mission Systems, applicable compliance requirements are captured in a single table. Compliance is documented for each applicable requirement resulting in an assigned compliance tier and a weighted compliance score for three categories of Mission System requirements, including Mission System, MASI, and Cyber categories. Individual MPU scores do not contribute to the Mission System summary scores, but the scores are reported

separately as part of a Mission System assessment report.

Mission System Adaptation

A Mission System may incorporate legacy elements by providing hardware or software adapters for the element. In this case, for a compliance assessment of a Mission System, the legacy element and its wrappers together act as a single MPU for assessing a score. Calculation of this score will depend upon the nature of the adapted element. For example, if the adaptation creates an OMS Subsystem, then an OMS compliance assessment may be appropriate for that element.

A Mission System may incorporate legacy subsystems by providing hardware or software adapters for the subsystem. The procuring authority determines how to represent the compliance of legacy subsystems and elements.

A Mission System can be AMS GRA-compliant even if it uses legacy subsystems that are not AMS GRA-compliant if approved by the procuring authority. An AMS GRA Mission System may contain MPUs with varying compliance scores. Likewise, a Mission System may also contain legacy subsystems that are not AMS GRA-compliant.

2.3.4 Compliance Notes for Interfaces

All interfaces associated with an MPU are identified in the MPU's interface table and diagram. MPU interface diagrams identify each applicable interface as Allowed/Required and define each interface's necessary Openness. MPU interface tables provide descriptions of each applicable interface.

The Allowed/Required interface type has the following meaning for the MPU:

- **Required Interface:** A set of interfaces (hardware or software) that must be implemented within AMS GRA.
- **Conditionally-Required Interface:** A set of interfaces (hardware or software) that must be implemented when applicable to the specific instantiation of AMS GRA.
- **Allowed Interface:** An optional set of interfaces (hardware or software) that may be implemented.

The interface Openness level adheres to the following definitions:

- **Common Interface:** Fully documented, non-proprietary interface (hardware and software) standardized by AMS GRA.

- **Open Interface:** Fully documented, non-proprietary interface (hardware and software) made available to AMS GRA community without restriction (the interface implementer can change the interface definition with the associated publication of documentation).
- **Exposed Interface:** Fully documented, proprietary interface (hardware and software) made available only with associated contractor agreements.

Exposed Interfaces may include elements that are Open and/or Common. However, if any element of an interface is Exposed, the interface is categorized as Exposed. Likewise, Open Interfaces may include elements that are Common, but if any element of the interface is Open, then the interface is categorized as Open. An exception to this rule is when a Common interface is designed to allow for Open Additions; as long as the entire required portion of the interface is Common, the interface is considered Common (not Open).

2.3.5 Compliance Notes for Developed Software

AMS GRA Services and OMS Services must be instrumented in a Development, Security, and Operations (DevSecOps) pipeline and all artifacts required by the MPU compliance requirements be provided as objective evidence of compliance. When MPUs consist of government accredited solutions, it is possible that policy, precedence, or operating procedures prohibit or prevent these MPUs from following the AMS GRA compliance assessment approach. For example, Security Modules (SMs) may contain a National Security Agency (NSA) Type 1 Encryptor. In this situation, appropriate government accreditation documentation may substitute for compliance to both the DevSecOps pipeline and any compliance artifacts corresponding to the MPU. In cases where documentation is not available, weighted compliance assessment may be used to help tailor the impact of such situations.

For MPUs containing microprocessor code that is not developed by the MPU developer (e.g. Commercial-Off-The-Shelf (COTS) open source, third-party libraries, or Government-Off-The-Shelf (GOTS)), MPU microprocessor compliance requirements may be waived.

An MPU that is delivered in a container is allowed to be removed from the container for execution. Services may also be delivered in a non-containerized format in addition to delivery in a containerized format. Note that GRA-MASI-021 and GRA-MASI-022 do not preclude a MASI from deploying non-containerized software. A procuring entity may remove MPU software from its container if appropriate for their use case. Suppliers may also deliver services in a non-containerized format to support such use cases, so long as they provide a containerized version.

2.4 Framework for Compliance Assessment

Different parties have different responsibilities for compliance assessment as outlined below:

- The **procuring authority** is ultimately responsible for assessing the compliance of a Mission System against AMS GRA.
- The **MSI** is responsible for compiling the assessment of architectural compliance of all MPUs that compose the Mission System, and providing an MPU compliance table to the procuring authority.
- The **MSI** is responsible for conducting a self-assessment of the overall Mission System compliance characteristics IAW the Mission System Volume. The **MSI** is also responsible for providing the results of this self-assessment, including substantiating documentation, to the procuring authority.
- Each **MPU provider** is responsible for conducting a self-assessment of MPU compliance characteristics IAW the compliance table for the MPU, and providing the results, with substantiating documentation, to the MSI or procuring authority.
- Certain MPU requirements target optional or conditionally-required capabilities that may not be present in all instantiations of a particular type of MPU. If a specific instantiation of an MPU does not include the targeted capability, then the requirement is not applicable for that specific instance. For example, if the Processor Chassis and Backplane (PCaB) has optional system management software capabilities, GRA-MPU-061 will apply. However, if the PCaB does not implement the optional system management interface, GRA-MPU-061 will not apply and should not be included in the compliance evaluation.

Figure 2.4.1 shows a simplified example of how two different MPUs receive different tier assessments and weighted compliance scores. Note that if two different procuring authorities select different (non-zero) weights, the weighted score may change from one assessment to the next although the tier will remain the same.

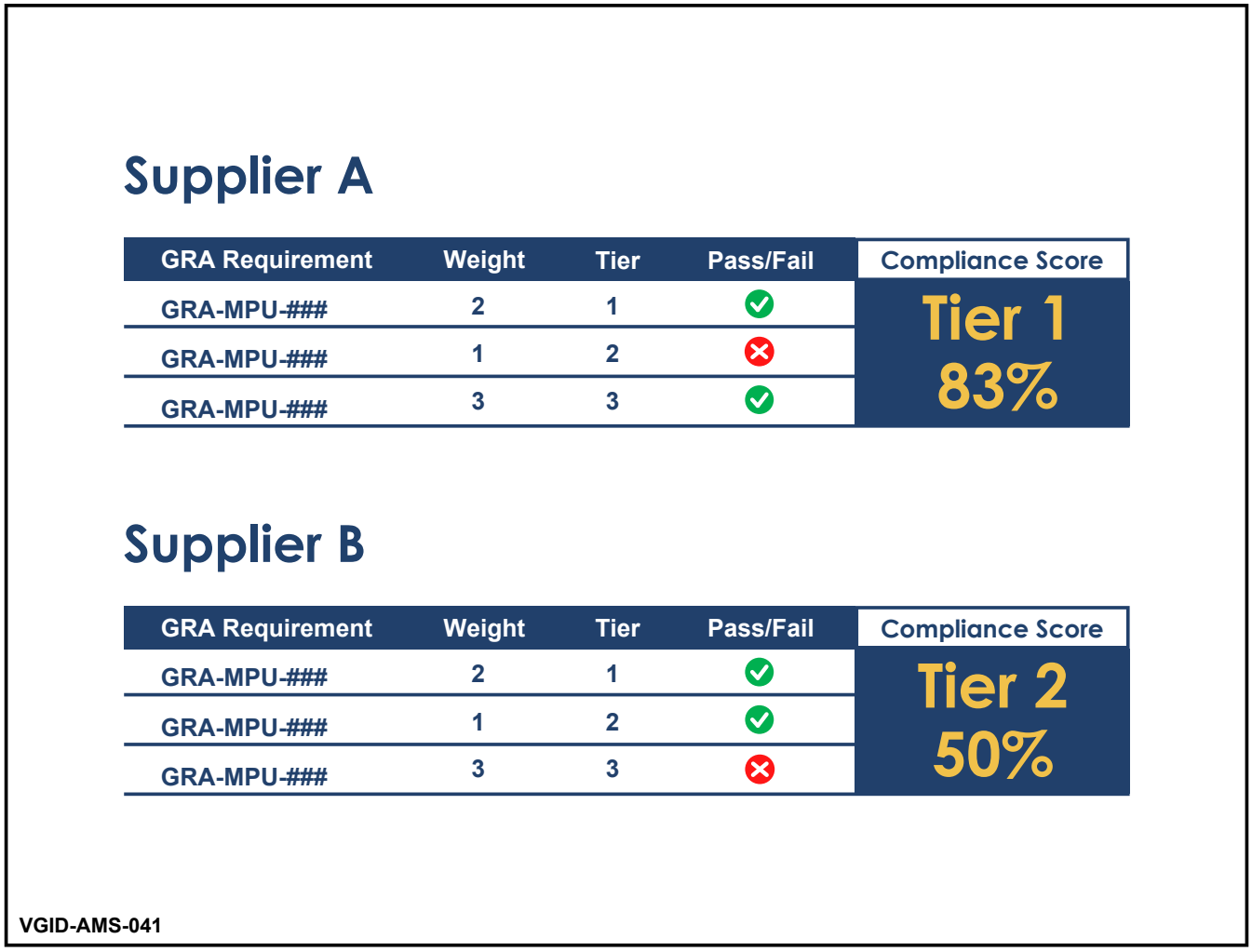


Figure 2.4.1: The AMS GRA Example Compliance Scorecard Compares Two Suppliers, A and B, Reflecting How Tiers and Scores Are Impacted by Passing/Failing Requirements.

The compliance assessment scorecard templates are provided to help adopters calculate compliance against AMS GRA requirements. The scorecard calculates both aspects of compliance - a weighted score and an achieved compliance tier.

Two scorecard templates are included, one for the MPU and DK requirements and the other for the Cyber, MASI, and Mission System requirements. Each requirement can be evaluated as pass, fail, or not applicable. The score-

card is set up to filter out non-applicable MPU types from the compliance and tiering evaluation. The compliance score is calculated based on the associated weight and the pass/fail status of the requirements. The tiering score is calculated based on **Figure 2.4.1**.

3.0 Standards

AMS GRA leverages existing standards and specifications. When standards are insufficient to meet mission needs, tailoring or modification is allowable. The delta implementation must include documented rationale and list impacts (e.g., consider 10-15 years of modified off-the-shelf). Any partial use or tailoring called out by AMS GRA supersedes required compliance to the original standard, specification, or protocol.

When a Mission System is under development, the procuring authority provides additional guidance for use, partial use, or tailoring of existing standards, specifications, and protocols.

In situations where the incorporation of a specific standard, specification, or protocol directly contradicts an AMS GRA architectural goal, the implementer documents their proposed exception and provides it, along with an explanation, to the procuring authority. The procuring authority then approves or denies their exception to the standard, specification, or protocol. Review of AMS GRA should occur at major AMS GRA document releases. Reviews should consider whether updates to AMS GRA are appropriate, or whether specific tailoring of referenced specific standards, specifications, or protocols should occur.

Examples of potential standards, specifications, or protocols for use by AMS GRA or the procuring authority include, but are not limited to OMS, Military Standards (MIL-STDs), VME bus International Trade Association (VITA), Common Open Architecture Radar Program Specification (COARPS), Future Airborne Capability Environment (FACE), Ethernet, and InfiniBand.

4.0 Assumed Procurement Approach

For new MPU development, expectations are that the procuring authority provides additional guidance beyond the basic compliance criteria. Examples of potential additional guidance are included below:

- For all MPU types, the procuring authority provides architecture guidance to ensure that the MPU integrates within the overall Mission System. In some situations, this may include detailed interfaces or operational behavior to ensure operation of the MPU in the overall architecture.
- For Level 2 and Level 3 MPUs, the procuring authority provides guidance to align with COTS standards (e.g., physical form factor, cooling strategy, environmental operating conditions, etc.).

MPUs that compose the Mission System must be compatible with these characteristics. During supplier selection, the MSI has discretion to evaluate MPU offerings based upon factors including MPU alignment with AMS GRA business and technical objectives, MPU architectural compliance, MPU commonality across the platform, and MPU compatibility with the platform architecture.

5.0 Compliance Requirements

Table 5.1.1, **Table 5.2.1**, and **Table 5.3.1** are the full set of compliance requirements for AMS GRA including Mission System requirements, MPU requirements, and DK requirements, respectively.

5.1 Mission System Requirements

Table 5.1.1 provides the requirements applicable to the Mission System as a whole.

Id	Text	Compliance Tier	Recommended Verification Method
GRA-CYAGE-002	A Cyber Agent shall be able to send and receive the messages defined in the Cyber Agent Messages section of the Cyber ICD.	Tier 1	Demonstration
GRA-CYAGE-003	A Cyber Agent shall execute actions at the direction of the Cyber Manager.	Tier 2	Demonstration
GRA-CYAGE-004	A Cyber Agent shall possess functionality and interfaces to execute actions defined within the Cyber Agent Messages section of the Cyber ICD.	Tier 2	Demonstration
GRA-CYAGE-005	A Cyber Agent shall communicate with a Cyber Manager using secure communication.	Tier 3	Analysis
GRA-CYMAN-002	A Cyber Manager shall be able to send and receive the messages defined in the Cyber Manager Messages of the Cyber ICD.	Tier 1	Demonstration
GRA-CYMAN-003	A Cyber Manager shall generate state, mode, and event detail logs.	Tier 1	Demonstration
GRA-CYMAN-004	A Cyber Manager shall be permitted to reconfigure enforcement policies of the Cyber Monitor at runtime using the Cyber Manager messages defined in the Cyber ICD.	Tier 2	Demonstration
GRA-CYMAN-005	A Cyber Manager shall orchestrate the collection of the data from Cyber Monitors.	Tier 2	Demonstration
GRA-CYMAN-006	A Cyber Manager shall dynamically reconfigure a Cyber Monitor using the Cyber Manager Messages defined in the Cyber ICD.	Tier 2	Demonstration
GRA-CYMAN-007	A Cyber Manager shall detect anomalous behavior indicative of cyber events within the Mission System.	Tier 2	Demonstration
GRA-CYMAN-008	A Cyber Manager, IAW its current policy configuration, shall orchestrate the reactions of a Cyber Agent.	Tier 2	Demonstration
GRA-CYMAN-009	A Cyber Manager shall communicate with a Cyber Monitor using secure communication.	Tier 1	Inspection
GRA-CYMAN-010	A Cyber Manager shall communicate with a Cyber Agent using secure communication.	Tier 3	Inspection
GRA-CYMON-002	A Cyber Monitor shall be able to send and receive the messages defined in the Cyber Monitor Messages section of the Cyber ICD.	Tier 1	Demonstration
GRA-CYMON-003	A Cyber Monitor shall possess functionality and interfaces to retrieve data defined within the Cyber Monitor Messages section of the Cyber ICD from MPUs.	Tier 2	Demonstration
GRA-CYMON-004	A Cyber Monitor shall be reconfigurable at runtime by the Cyber Manager using the Cyber Monitor messages defined in the Cyber ICD.	Tier 3	Demonstration
GRA-CYMON-005	A Cyber Monitor shall communicate with a Cyber Manager using secure communication.	Tier 1	Inspection
GRA-MASI-001	A MASI shall be able to start mission-specific and Mission-Agnostic Services.	Tier 1	Demonstration
GRA-MASI-002	A MASI shall be able to stop mission-specific and Mission-Agnostic Services.	Tier 1	Demonstration

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MASI-003	A MASI shall monitor the Management Interface for reported health and status of AMS GRA-compliant hardware.	Tier 1	Analysis
GRA-MASI-006	A MASI shall monitor fault information reported by AMS GRA-compliant MPUs.	Tier 1	Inspection
GRA-MASI-008	A MASI shall consume all available BIT information reported by AMS GRA-compliant hardware.	Tier 1	Analysis
GRA-MASI-010	A MASI shall provide a mechanism to aggregate disparate fault, health, and status data.	Tier 2	Inspection
GRA-MASI-011	A MASI shall provide a mechanism to store fault, health, and status data.	Tier 2	Inspection
GRA-MASI-012	A MASI shall be able to respond to fault information reported by AMS GRA-compliant MPUs.	Tier 2	Analysis
GRA-MASI-013	A MASI shall provide an interface to access fault, health, and status data.	Tier 2	Inspection
GRA-MASI-015	A MASI shall monitor the health and status reporting of all deployed AMS GRA and OMS Services.	Tier 1	Inspection
GRA-MASI-016	A MASI shall monitor compute resource information (e.g., utilization, availability, capability, bandwidth, flash, RAM).	Tier 2	Inspection
GRA-MASI-017	A MASI shall provide a mechanism to allow services to access all available software and firmware configuration status and versioning information for software and firmware stored and deployed on the AMS GRA Mission System.	Tier 3	Demonstration
GRA-MASI-020	A MASI shall provide a mechanism to consolidate duplicate fault data.	Tier 2	Inspection
GRA-MASI-021	A MASI shall include an OCI-compliant Orchestration Engine.	Tier 2	Inspection
GRA-MASI-022	A MASI shall deploy OCI-compliant containerized software utilizing a container framework.	Tier 1	Demonstration
GRA-MASI-023	A MASI shall provide access to compute resource information.	Tier 2	Inspection
GRA-MASI-024	A MASI shall facilitate endpoint connections between services.	Tier 1	Demonstration
GRA-MASI-025	A MASI shall provide a mechanism to allow services to access Local Function information.	Tier 2	Inspection
GRA-MASI-027	A MASI shall register the services available for deployment.	Tier 1	
GRA-MASI-028	A MASI shall report the state of MASI	Tier 1	
GRA-MASI-029	A MASI shall support deployment of a minimum of two instances of the same Service.	Tier 1	Test
GRA-MASI-030	A MASI shall isolate AMS GRA and OMS Service software faults from other software MPUs.	Tier 1	Test
GRA-MASI-031	A MASI shall assign an MPUs' deployment targets and resources in accordance with that MPUs's delivered service.yml file.	Tier 1	Test

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MASI-032	A MASI shall provide an implementation of the AMS Common Transport API.	Tier 1	
GRA-MASI-033	A MASI shall report the status of all SW MPUs available in the system in UCI format.	Tier 2	Test
GRA-MASI-034	A MASI shall report compute resource information in UCI format if requested in UCI format by an MPU within the system.	Tier 2	Test
GRA-MS-001	A Mission System shall implement a single physical ABB that supports the AMS GRA MPU interfaces that use the ABB.	Tier 1	Inspection
GRA-MS-002	A Mission System shall implement a single physical ASB IAW OMS standard.	Tier 1	Inspection
GRA-MS-003	A Mission System shall implement network accessible Data Storage.	Tier 1	Inspection
GRA-MS-004	A Mission System shall implement a documented architecture that supports the distribution of time to support platform-implemented MPUs including PTP, DGT, and/or OMS legacy timing.	Tier 1	Inspection
GRA-MS-005	A Mission System shall implement access and information flow isolation between safety critical and non-safety critical processing.	Tier 1	Analysis
GRA-MS-006	A Mission System that includes SMS shall implement the WSI as defined in the Armament Volume.	Tier 1	Inspection
GRA-MS-007	A Mission System shall implement an API on DPPs, as applicable, to allow AMS GRA Services and OMS Services to access OMS Special Signals.	Tier 2	Inspection
GRA-MS-008	A Mission System shall implement an API on DPPs, as applicable, to allow AMS GRA Services and OMS Services to access OMS platform-specific Elements.	Tier 2	Inspection
GRA-MS-009	A Mission System shall implement an OMS Critical Abstraction Layer on DPPs to allow AMS GRA Services and OMS Services to access to UCI messaging.	Tier 1	Inspection
GRA-MS-010	A Mission System shall provide an AMS GRA-compliant MASI implementation.	Tier 1	Inspection
GRA-MS-013	A Mission System that segregates data into multiple security domains shall utilize an approved Transfer CDS to pass data between security domains.	Tier 3	Analysis
GRA-MS-014	A Mission System utilizing an MLS bus shall enforce logical segregation of security domains by utilizing an approved MLS CDS at each bus interface.	Tier 3	Analysis
GRA-MS-020	A Mission System shall preserve availability of a trusted copy of Mission System software, firmware, and configuration files during a failed reprogramming attempt.	Tier 3	Test
GRA-MS-023	A Mission System shall make available all MEL Implementations supplied by MFA providers to any AMS GRA Service to be loaded.	Tier 1	Analysis
GRA-MS-024	A Mission System shall provide connectivity to the ASB from any DPP in the CCF.	Tier 1	Analysis
GRA-MS-025	A Mission System shall provide connectivity to the ABB from any DPP performing MFP layer processing within the CCF.	Tier 1	Analysis

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MS-026	A Mission System shall make available to MASI the locality between DPPs, and DPPs to MFAs to inform the deployment and recovery strategy.	Tier 2	Inspection
GRA-MS-027	A Mission System shall provide the capability of all AMS GRA Services to be deployed to any ABB-connected compatible DPP.	Tier 1	Analysis
GRA-MS-028	A Mission System employing MLS logical separation on an MLS bus shall provide interoperable authentication and integrity mechanisms consistent for communicating endpoints in an approved manner.	Tier 3	Analysis
GRA-MS-029	The Mission System shall provide open documentation for the incorporated MLS solution regarding programming of security policies and switch configuration if multiple security domains or security policies are present in support of approved certification.	Tier 3	Inspection
GRA-MS-030	A Mission System shall include non-proprietary documentation for interfaces operating at multiple security domains.	Tier 3	Inspection
GRA-MS-031	A Mission System shall include MLS interface documentation with descriptions of boundaries and allowable data flows between multiple interconnected security domains.	Tier 3	Inspection
GRA-MS-032	A Mission System shall include interface documentation for hardware requirements and interface design details when multiple interconnected security domains are present within the Mission System.	Tier 3	Inspection
GRA-MS-033	A Mission System with classified data shall encrypt all classified DaR using approved cryptographic algorithms.	Tier 1	Inspection
GRA-MS-034	A Mission System shall include a model consistent with MBSE content contained in the AMS GRA SK release with which the Mission System is compliant.	Tier 2	Inspection
GRA-MS-035	A Mission System shall use approved cryptographic algorithms to authenticate and verify the integrity of all Mission System data, software and firmware prior to updating, loading, and executing.	Tier 3	Inspection
GRA-MS-036	The Mission System documentation shall document the implementation of the approved cryptographic authentication and verification algorithms for all loadable Mission System data, software and firmware prior to updating, loading, and executing.	Tier 3	Inspection
GRA-MS-037	A Mission System shall generate audit logs for security relevant events to include generation of audit logs, response to audit failure, protection of audit information and enforcement of non repudiation.	Tier 2	Inspection
GRA-MS-041	A Mission System shall provide storage for Local Functions to be accessible to the associated MFA using networked Data Storage, non-volatile MFA local storage, or non-volatile DPP local storage.	Tier 2	Inspection

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MS-045	A Mission System shall include SAST results for all microprocessor software source code in the Mission System, IAW the Mission System documentation template.	Tier 1	Inspection
GRA-MS-046	A Mission System shall include DAST results for all microprocessor software in the Mission System, IAW the Mission System documentation template.	Tier 1	Inspection
GRA-MS-049	A Mission System with CPI in non-volatile memory shall prevent recovery of CPI and protective keys in the event of a loss of Mission Platform.	Tier 1	Analysis
GRA-MS-050	A Mission System shall provide interface connectivity for Required MPU interfaces.	Tier 1	Inspection
GRA-MS-051	A Mission System shall provide configurability for audit of security relevant events.	Tier 2	Inspection
GRA-MS-052	A Mission System shall securely deploy software containing CPI.	Tier 1	Demonstration
GRA-MS-054	A Mission System shall include documentation on the deployment and recovery strategy for its software elements.	Tier 1	Inspection
GRA-MS-055	A Mission System shall include non-proprietary machine readable documentation for the security audit log format.	Tier 2	Inspection
GRA-MS-056	A Mission System with MFAs shall provide an implementation of the Resource Management Framework for management of sensor resources.	Tier 1	Inspection
GRA-MS-057	The Mission System shall provide a RoT to Mission System MPUs.	Tier 2	Analysis
GRA-MS-059	A Mission System shall verify the integrity of a hardware MPU over a secure communications channel prior to enabling the transfer of software, mission data, or associated Key Material to/from that MPU.	Tier 2	Demonstration
GRA-MS-060	The Mission System shall orchestrate secure communications for ABB and ASB connected endpoints.	Tier 1	Analysis
GRA-MS-061	The Mission System shall authorize access to endpoints on a per-session basis.	Tier 2	Analysis
GRA-MS-062	The Mission System shall establish secure communications between endpoints using NSA-approved cryptographic methods.	Tier 2	Inspection
GRA-MS-063	The Mission System shall implement dynamic authorization for secure communications.	Tier 3	Analysis
GRA-MS-064	The Mission System shall dynamically revoke session authentication when directed by an authorized source.	Tier 3	Analysis
GRA-MS-065	The Mission System shall only allow authorized communication channels IAW a file identified and documented by the Mission System.	Tier 2	Demonstration
GRA-MS-066	The Mission System shall mutually authenticate all endpoints.	Tier 2	Analysis
GRA-MS-067	The Mission System shall provide a KMP.	Tier 2	Inspection

Id	Text	Compliance Tier	Recommended Verification Method
GRA-MS-068	The Mission System shall protect security functions from unauthorized access or modification.	Tier 1	Demonstration
GRA-MS-069	The Mission System shall protect security relevant data from unauthorized access or modification.	Tier 1	Demonstration
GRA-MS-071	The Mission System shall use secure communications to transfer classified key material between Mission System endpoints.	Tier 1	Demonstration
GRA-MS-072	The Mission System shall use secure communications to transfer classified software between Mission System endpoints.	Tier 1	Demonstration
GRA-MS-073	The Mission System shall use the MEL Implementation supplied by the MFA provider to establish the interface between an AMS GRA Service and the MFA.	Tier 1	Demonstration
GRA-MS-074	A Mission System shall restrict access to OMS Special Signals through DPP APIs to non-message based signals that cannot be feasibly implemented using AMS GRA- defined common or open forms of communication.	Tier 2	Inspection
GRA-MS-075	A Mission System shall provide a Critical Program Information (Critical Program Information (CPI)) analysis, to include Consequence of Compromise (Consequence of Compromise (CofC)), in accordance with the Department of Defense Anti-Tamper Engineering Framework.	Tier 1	Inspection
GRA-MS-076	A Mission System shall instantiate the Cyber Agent.	Tier 3	Demonstration
GRA-MS-077	A Mission System shall instantiate the Cyber Monitor.	Tier 3	Demonstration
GRA-MS-078	A Mission System shall instantiate the Cyber Manager.	Tier 3	Demonstration
GRA-MS-079	A Mission System shall publish the Service health and status messages on the ABB to the ASB	Tier 1	Inspection

Table 5.1.1: A Mission System's Adherence to the Compliance Requirements Determines Its Compliance Score.

5.2 Minimum Procurable Unit (MPU) Requirements

Not all MPU requirements are applicable to each MPU. The applicability of the MPU requirements are given in **Table 5.2.1**. The requirements for each individual MPU are also given in their respective Volume.

The GRA-DK referenced in **Table 5.2.1** can be found in **Table 5.3.1**.

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-001	An MPU shall be documented IAW the corresponding AMS GRA compliance documentation template or contract template, as applicable.	Tier 1	Inspection	AMS GRA Service CDS CMM Cryptographic Unit Data Storage DPP Local Function MFA PCaB Switch	
GRA-MPU-002	An MPU shall be documented IAW the OMS Service Contract Template and OMS Service Contract Addendum.	Tier 1	Inspection	OMS Service	
GRA-MPU-003	An MPU targeting deployment on a GPP DPP shall be restricted to using the POSIX API for OS system calls.	Tier 1	Inspection	AMS GRA Service OMS Service	
GRA-MPU-004	An MPU shall meet all of the OMS Tier 1 requirements for an OMS Service.	Tier 1	Inspection	OMS Service	
GRA-MPU-005	An MPU with CPI on a Trusted Ops platform shall cryptographically protect CPI using an open implementation.	Tier 3	Analysis	AMS GRA Service CMM Data Storage MFA OMS Service Switch	
GRA-MPU-006	An MPU that processes or decrypts CPI on a built-for-export platform shall implement the OMS Standard.	Tier 2	Analysis	AMS GRA Service CMM Data Storage MFA OMS Service Switch	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-007	An MPU that implements the Time interface shall use one or more of PTP, DGT, and OMS legacy timing.	Tier 1	Inspection	CMM MFA	
GRA-MPU-008	An MPU shall fully implement the required interface elements defined by the Common MEL API corresponding to its MFA type, but may include Open additions.	Tier 1	Inspection	MFA	
GRA-MPU-009	An MPU shall provide one or more software libraries implementing the Common MEL API corresponding to its MFA Type.	Tier 1	Inspection	MFA	
GRA-MPU-011	An MPU that hosts Level 1 MPUs shall provide a DK IAW DK requirements.	Tier 1	Inspection	DPP MFA	
GRA-MPU-012	An MPU that hosts MFA OEM Functions and Local Functions shall use either MFA Compute hardware outside the CCF, DPP MPUs inside the CCF, or a combination of both.	Tier 3	Inspection	MFA	
GRA-MPU-013	An MPU's MEL implementation shall use the ABB for all information exchanges.	Tier 1	Demonstration	MFA	
GRA-MPU-014	An MPU shall implement a PTP client compatible with IEEE 1588-2008.	Tier 1	Inspection	CDS Cryptographic Unit DPP	
GRA-MPU-015	An MPU shall support PTP time compatible with IEEE 1588-2008	Tier 1	Inspection	Switch	MPU passes PTP packets as normal traffic
GRA-MPU-017	An MPU shall support loading, updating, and zeroizing keys and certificates using an approved cryptographic fill interface.	Tier 1	Inspection	Cryptographic Unit	
GRA-MPU-018	An MPU shall implement interfaces according to the CDS IDD.	Tier 1	Inspection	CDS	
GRA-MPU-019	An MPU that implements an OMS interface shall implement the interface IAW OMS requirements.	Tier 1	Inspection	AMS GRA Service CMM	
GRA-MPU-020	An MPU shall report a normative service status message as defined by OMS.	Tier 1	Demonstration	AMS GRA Service	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-021	A generated MPU Status shall be available on the ASB Critical Abstraction Layer.	Tier 1	Demonstration	AMS GRA Service CMM	
GRA-MPU-022	An MPU shall use the MEL for all information exchanges between AMS GRA Services and MFAs.	Tier 1	Inspection	AMS GRA Service MFA	
GRA-MPU-023	An MPU shall report status to the MFA IAW the DK.	Tier 1	Demonstration	Local Function	
GRA-MPU-024	An MPU that interacts with the CDS or Cryptographic Unit shall use the CDS IDD and **CU!** IDD respectively for all exchanges with the Cryptographic Unit or CDS.	Tier 1	Inspection	AMS GRA Service CMM DPP MFA OMS Service	
GRA-MPU-025	An MPU shall report faults with the associated fault status.	Tier 1	Inspection	All	Off-nominal behavior detailed in compliance documentation.
GRA-MPU-026	An MPU shall implement Required interfaces specified IAW the AMS GRA Model.	Tier 1	Inspection	All	
GRA-MPU-027	An MPU shall implement all interfaces with the level of Openness specified IAW the AMS GRA Model.	Tier 1	Analysis	All	
GRA-MPU-028	An MPU shall provide approved cryptographic functions, to include (but not limited to): key management (e.g., generation, storage/backup, distribution, update/loading), cryptologic transformation (e.g. encryption and decryption), and digital signature generation and verification.	Tier 1	Analysis	Cryptographic Unit	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-029	An MPU shall implement only the interfaces in the corresponding MPU interface table.	Tier 2	Inspection	AMS GRA Service CDS CMM Cryptographic Unit Data Storage DPP Local Function MFA PCaB Switch	
GRA-MPU-030	An MPU shall document interface openness IAW the Compliance Assessment Approach.	Tier 1	Inspection	All	
GRA-MPU-031	An MPU shall include a model compatible with MBSE content contained in the AMS GRA SK release with which the MPU is compliant.	Tier 2	Inspection	All	
GRA-MPU-032	An MPU shall include a model that identifies all external MPU interfaces.	Tier 2	Inspection	All	
GRA-MPU-034	An MPU that supports third-party development of VAs shall provide a VA Development Tool.	Tier 1	Inspection	MFA	
GRA-MPU-035	An MPU that supports Aperture Virtualization and third-party development of Local Function using the Local Function DK shall provide a VA Development Tool capable of associating the Local Function with a VA.	Tier 1	Inspection	MFA	
GRA-MPU-036	An MPU that is capable of hosting Local Functions shall retrieve stored Local Functions from storage designated by the Mission System.	Tier 2	Inspection	MFA	
GRA-MPU-037	An MPU shall use, at a minimum, NFS as a protocol for the Data interface.	Tier 1	Inspection	Data Storage	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-038	An MPU shall use open, commercial standards for the Management interface.	Tier 1	Inspection	Data Storage DPP PCaB Switch	
GRA-MPU-039	An MPU shall provide a CLASSIFICATION.md file defining security classification of included file contents.	Tier 2	Inspection	AMS GRA Service CDS CMM Cryptographic Unit Local Function MFA OMS Service	
GRA-MPU-040	An MPU shall provide a DATARIGHTS.md file defining data rights of included file contents.	Tier 2	Inspection	AMS GRA Service CDS CMM Cryptographic Unit Local Function MFA OMS Service	
GRA-MPU-041	An MPU shall document SAST results for all microprocessor software source code in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	All	
GRA-MPU-042	An MPU shall document DAST results for all microprocessor software in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	All	
GRA-MPU-043	An MPU hosting Local Functions shall provide the capability in the MFA Compute Interface to obtain current MFA Time.	Tier 1	Inspection	MFA	
GRA-MPU-044	An MPU hosting Local Functions shall provide MFA Time for digitized sample data in its MFA Internal Data Interface.	Tier 1	Inspection	MFA	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-045	An MPU hosting Local Functions shall provide the capability to convert an MFA Time to a universal time reference.	Tier 1	Inspection	MFA	
GRA-MPU-046	An MPU shall provide automated tests of non-COTS microprocessor software with at least 50% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 1	Demonstration	All	Only applicable to MPUs that include non-COTS microprocessor code.
GRA-MPU-047	An MPU with microprocessor code (excluding third-party COTS or open source libraries) shall only use relative file paths in the execution environment.	Tier 2	Inspection	AMS GRA Service CDS CMM Cryptographic Unit Data Storage OMS Service	
GRA-MPU-048	An MPU with microprocessor code (excluding third-party COTS or open source libraries) shall only use relative file paths in the build environment.	Tier 2	Inspection	AMS GRA Service CDS CMM Cryptographic Unit Data Storage OMS Service	
GRA-MPU-049	An MPU that implements a scheduling function shall manage the timeline utilization of the MPU.	Tier 1	Demonstration	MFA	
GRA-MPU-050	An MPU with a scheduling function, that instantiates VAs, shall manage the timeline utilization of the VAs.	Tier 1	Demonstration	MFA	
GRA-MPU-051	An MPU shall include a software bill of materials in an open and non-proprietary format.	Tier 1	Inspection	AMS GRA Service Local Function OMS Service	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-053	An MPU's DPs shall each contain content only from that MPU.	Tier 3	Inspection	AMS GRA Service OMS Service	
GRA-MPU-057	An MPU's DP that targets a GPP Processing Component shall consist of a single OCI-compliant container.	Tier 1	Demonstration	AMS GRA Service OMS Service	
GRA-MPU-058	An MPU shall encrypt classified DaR using NSA-approved algorithms.	Tier 1	Analysis	AMS GRA Service CMM Cryptographic Unit Data Storage DPP MFA OMS Service Switch	
GRA-MPU-059	An MPU that decrypts and/or encrypts classified data and/or CPI shall only store classified plain text keys in volatile memory.	Tier 1	Inspection	AMS GRA Service CMM Cryptographic Unit Data Storage DPP Local Function MFA OMS Service PCaB Switch	
GRA-MPU-060	An MPU's write-protected memory shall be modifiable through a hardware element in addition to software.	Tier 1	Analysis	CDS CMM Cryptographic Unit DPP MFA	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-061	An MPU shall ensure the authenticity and integrity of MPU software and firmware during MPU initialization.	Tier 1	Inspection	CDS CMM Cryptographic Unit Data Storage DPP MFA PCaB Switch	
GRA-MPU-062	An MPU shall support integrity verification to the Mission System over a secure communication channel.	Tier 1	Demonstration	CDS CMM Cryptographic Unit Data Storage DPP MFA Switch	
GRA-MPU-063	An MPU shall implement secure communications using open standards.	Tier 1	Analysis	CDS CMM Cryptographic Unit Data Storage DPP MFA Switch	
GRA-MPU-064	An MPU shall use NSA-approved algorithms to authenticate and verify the integrity of a software load when received over a non-secure communication channel.	Tier 1	Demonstration	CDS CMM Cryptographic Unit Data Storage DPP MFA PCaB Switch	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-065	An MPU shall use NSA-approved algorithms to authenticate and verify the integrity of software locally stored at rest, not including bootloaders; prior to updating, loading, and executing.	Tier 1	Inspection	CDS CMM Cryptographic Unit Data Storage DPP MFA PCaB Switch	
GRA-MPU-066	An MPU shall support the use of a Mission System-provided RoT.	Tier 1	Demonstration	All	
GRA-MPU-067	An MPU shall publish details of security relevant events in an open format.	Tier 1	Demonstration	CDS Cryptographic Unit Data Storage DPP MFA Switch	
GRA-MPU-068	An MPU shall boot from write-protected memory.	Tier 1	Demonstration	CDS Cryptographic Unit DPP MFA	
GRA-MPU-069	An MPU shall format MEL data IAW the MEL IDD corresponding to the MFA Type. Open additions are permitted.	Tier 1	Demonstration	AMS GRA Service MFA	
GRA-MPU-070	An MPU using a MEL API Implementation to interact with an MFA shall use the interface elements defined by the Common MEL API corresponding to the MFA Type with Open Additions to that MEL.	Tier 1	Demonstration	AMS GRA Service	
GRA-MPU-071	An MPU shall be built, packaged, and delivered to persistent storage via automated pipeline.	Tier 1	Inspection	AMS GRA Service OMS Service	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-072	An MLS Transfer CDS shall manage security and access policies IAW the approval authority for logical separation and transfer of data between security domains.	Tier 1	Inspection	CDS	
GRA-MPU-073	An MLS Transfer CDS shall enforce security and access policies IAW the approval authority for logical separation and transfer of data between security domains.	Tier 1	Inspection	CDS	
GRA-MPU-074	An MPU that processes or decrypts CPI on a built-for-export platform shall fully implement OMS Standard.	Tier 3	Analysis	AMS GRA Service CMM Data Storage MFA OMS Service Switch	
GRA-MPU-075	An MPU shall implement the required interface elements defined by the Common MEL API corresponding to its MFA Type, and does not include Open additions.	Tier 2	Inspection	MFA	
GRA-MPU-076	An MPU shall implement an optionally-enabled PTP transparent clock or PTP boundary clock functionality with software-based timestamp generation compatible with IEEE 1588-2008.	Tier 2	Inspection	Switch	
GRA-MPU-077	An MPU shall provide automated tests of non-COTS microprocessor software with at least 80% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 3	Demonstration	All	
GRA-MPU-078	An MPU model shall be capable of automated verification with AMS GRA models.	Tier 3	Inspection	All	
GRA-MPU-079	An MPU shall include a model that provides definition of all external MPU interfaces.	Tier 3	Inspection	All	
GRA-MPU-080	An MPU shall format MEL data IAW the MEL IDD corresponding to the MFA Type.	Tier 2	Demonstration	AMS GRA Service MFA	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-081	An MPU using a MEL API Implementation to interact with an MFA shall use only the interface elements defined by the Common MEL API corresponding to the MFA Type.	Tier 2	Demonstration	AMS GRA Service	
GRA-MPU-082	An MPU shall pass 70% or more of applicable jobs in a AMS GRA compliant pipeline.	Tier 2	Inspection	AMS GRA Service OMS Service	
GRA-MPU-083	An MPU shall pass 100% of applicable jobs in a AMS GRA compliant pipeline.	Tier 3	Inspection	AMS GRA Service OMS Service	
GRA-MPU-084	An MPU that implements the Time interface shall use one or more of PTP, and OMS legacy timing.	Tier 1	Inspection	Data Storage PCaB	
GRA-MPU-085	An MPU shall report faults with the associated fault status using the applicable interface. (See notes).	Tier 2	Inspection	All	Cryptographic Unit MPUss and CDS MPUss report faults as defined in the **CU!** IDD and CDS IDD respectively. Local Function MPUs report faults as defined in Local Function DK. MFA MPUs report faults as defined in the applicable MEL IDD. AMS GRA Service, CMM, DPP, Data Storage, Switch, OMS Service, and PCaB MPUs report faults using a software interface in UCI format.
GRA-MPU-086	An MPU shall use the CMM IDD for all information exchanges between the MPU and the CMM.	Tier 1	Inspection	AMS GRA Service MFA OMS Service	
GRA-MPU-087	An MPU shall implement an optionally-enabled PTP transparent clock or PTP boundary clock functionality with hardware-based timestamp generation compatible with IEEE 1588-2008.	Tier 3	Inspection	Switch	

Id	Text	Compliance Tier	Recommended Verification Method	Applicable MPUs	Notes
GRA-MPU-088	An MPU shall report health and status on its Management Interface.	Tier 1	Demonstration	CDS Cryptographic Unit Data Storage DPP Switch	
GRA-MPU-089	An MPU shall implement interfaces according to the Cryptographic Unit IDD.	Tier 1	Inspection	Cryptographic Unit	
GRA-MPU-090	An MPU shall provide a configuration file named service.yml with contents in accordance with the Service MPU Configuration Template YAML file.	Tier 1	Inspection	AMS GRA Service OMS Service	
GRA-MPU-091	An MPU shall report BIT information on its Management Interface.	Tier 1	Inspection	CDS CMM Cryptographic Unit Data Storage DPP PCaB Switch	
GRA-MPU-092	An MPU shall include a software bill of materials in an open and non-proprietary format.	Tier 2	Inspection	CDS CMM Cryptographic Unit DPP MFA	

Table 5.2.1: An MPU's Adherence to the Compliance Requirements Determines Its Compliance Score.

5.3 Development Kit Requirements

Table 5.3.1 provides DK requirements that apply to DPPs and MFA MPUs. These requirements are part of the authoritative AMS GRA compliance assessment. This table indicates the processor types (e.g. GPP, GPU, and FPGA) to which each requirement applies in order to assist the respective DK definition activity. In some cases, the MSI may replace the content in a requirement originally provided by the DPP or MFA MPU provider. The column titled MSI indicates the requirements for which the MSI may serve as an alternate provider of the content.

There are three patterns that refine some of the DK requirement definitions in the table:

- First, some requirements exist in pairs such that one requirement discusses a need for a commodity item while the other requirement presents the need for a custom tool.
- Second, and related to the first, some requirements only request the identification of certain DK content while others require the actual provision of the content. For example, a DK provider that uses a commodity item with a license would identify the item so a DK user could procure the item and acquire a license. A DK provider that uses a custom item would need to provide the item to the DK user.
- Third, some requirements include the phrase “if required for development.” These requirements may or may not apply to the compliance assessment depending on the DK provider’s implementation of the DK. If the DK implementation enables development without the need for a requirement that is marked in this way, then the requirement does not apply.

Id	Text	Compliance Tier	Recommended Verification Method	GPP	GPU	FPGA	MSI	Notes
GRA-DK-001	A DK shall provide a developer guide and support documentation, including a Software Bill of Materials, for a developer to use the DK and integrate within a DevSecOps tool chain.	Tier 1	Inspection	Yes	Yes	Yes	No	
GRA-DK-002	A DK shall include a build environment with pre-installed build infrastructure (where licensing/redistribution restrictions allow) if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	Applies only to DPP Build Infrastructure could consist of virtual machines and/or containers
GRA-DK-003	A DK shall include a development environment with libraries and tools pre-installed (where licensing/redistribution restrictions allow) if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	Development environment could consist of virtual machines or containers
GRA-DK-004	A DK shall provide simulators if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-005	A DK shall provide emulators if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-006	A DK shall provide software configuration management support to facilitate integration of software into the DevSecOps environment.	Tier 1	Inspection	Yes	Yes	Yes	Yes	

Id	Text	Compliance Tier	Recommended Verification Method	GPP	GPU	FPGA	MSI	Notes
GRA-DK-007	A DK shall provide build infrastructure configuration to configure the build environment.	Tier 1	Inspection	Yes	Yes	No	Yes	Infrastructure configuration could be provided either as documentation or as code Updated to "build infrastructure" and "build environment". The term "build environment" was used to allow developers to define their development workspace.
GRA-DK-008	A DK shall provide a Service Framework Reference Implementation for the design and implementation of a Skill.	Tier 3	Inspection	Yes	Yes	Yes	No	
GRA-DK-009	A DK shall provide a container framework if required for development.	Tier 1	Inspection	Yes	No	No	Yes	
GRA-DK-010	A DK shall provide a unit testing framework for a DPP. This framework enables the creation of unit tests for Level 1 MPUs.	Tier 1	Inspection	Yes	Yes	Yes	No	
GRA-DK-011	A DK shall provide target support packages to configure the specific processing resource.	Tier 1	Inspection	Yes	Yes	Yes	No	
GRA-DK-012	A DK shall identify specific versions of commodity build support utilities in a machine-readable format.	Tier 1	Inspection	Yes	Yes	Yes	Yes	An SBOM is a means to achieve machine readability.
GRA-DK-013	A DK shall provide custom build support utilities if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-014	A DK shall provide documentation of compiler flags and/or configuration properties if required for development.	Tier 1	Inspection	Yes	Yes	No	Yes	

Id	Text	Compliance Tier	Recommended Verification Method	GPP	GPU	FPGA	MSI	Notes
GRA-DK-015	A DK shall provide test tools for applicable Level 1 MPUs.	Tier 1	Inspection	Yes	Yes	Yes	Yes	The level 2 DPP MPU provides a DK for applicable level 1 MPUs such as AMS or OMS services.
GRA-DK-016	A DK shall identify specific versions of commodity tools if required for development in a machine-readable format.	Tier 1	Inspection	Yes	Yes	Yes	Yes	An SBOM is a means to achieve machine readability.
GRA-DK-017	A DK shall provide custom tools if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-018	A DK shall identify software integrity tools for a DPP.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-019	A DK shall provide custom debug tools if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-020	A DK shall identify specific versions of commodity packaging scripts and/or tools in a machine-readable format.	Tier 1	Inspection	Yes	Yes	Yes	Yes	An SBOM is a means to achieve machine readability.
GRA-DK-021	A DK shall provide custom packaging scripts and/or tools if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-022	A DK shall identify DASTs if required for development.	Tier 3	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-023	A DK shall identify SASTs and associated configurations using IaC if required for development.	Tier 3	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-024	A DK shall identify specific versions of commodity release packaging in a machine-readable format.	Tier 1	Inspection	Yes	Yes	Yes	Yes	An SBOM is a means to achieve machine readability.
GRA-DK-025	A DK shall provide custom release packaging if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-026	A DK shall identify specific versions of commodity dependent documentation and/or libraries in a machine-readable format if required for development.	Tier 1	Inspection	Yes	Yes	Yes	Yes	An SBOM is a means to achieve machine readability.

Id	Text	Compliance Tier	Recommended Verification Method	GPP	GPU	FPGA	MSI	Notes
GRA-DK-027	A DK shall provide specialty infrastructure libraries if required for development.	Tier 1	Inspection	Yes	Yes	No	Yes	
GRA-DK-028	A DK shall provide MASI elements whose behavior is required for development.	Tier 1	Inspection	Yes	No	No	Yes	
GRA-DK-029	A DK shall provide the processing resource tool configuration if required for development in a machine-readable format.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-030	A DK shall provide FPGA intellectual property Blocks if required for development.	Tier 1	Inspection	No	No	Yes	Yes	
GRA-DK-032	When used for DPP development, the DK shall provide a DPP API implementation required for development.	Tier 1	Inspection	Yes	No	No	Yes	Machine readable is supposed to be an SBOM
GRA-DK-033	A DK shall provide CI/CD support configuration files necessary to integrate DK components into a DevSecOps pipeline.	Tier 1	Inspection	Yes	Yes	No	Yes	
GRA-DK-034	A DK shall provide available OEM Function interfaces when used for Local Function development.	Tier 1	Inspection	Yes	Yes	Yes	No	
GRA-DK-035	A DK shall include a containerized build environment with pre-installed build infrastructure (where licensing/redistribution restrictions allow) if required for development.	Tier 2	Inspection	Yes	Yes	Yes	Yes	Applies only to DPP Build Infrastructure must consist of containers only
GRA-DK-036	A DK shall include a containerized build environment with pre-installed build infrastructure (where licensing/redistribution restrictions allow) if required for development.	Tier 3	Inspection	Yes	Yes	Yes	Yes	Applies only to MFA Build Infrastructure must consist of containers only
GRA-DK-037	A DK development environment shall include debugging tools and libraries created by the MPU provider to streamline development for the target MPU.	Tier 3	Inspection	Yes	No	No	No	
GRA-DK-038	A DK shall provide build infrastructure configuration as IaC to configure the build environment.	Tier 3	Inspection	Yes	Yes	No	Yes	

Id	Text	Compli-ance Tier	Recommended Verification Method	GPP	GPU	FPGA	MSI	Notes
GRA-DK-039	A DK shall provide a unit testing framework for an MFA. This framework enables the creation of unit tests for Level 1 MPUs.	Tier 3	Inspection	Yes	Yes	Yes	No	
GRA-DK-040	A DK shall provide documentation (as `Cmake` or equivalent project settings as code) of compiler flags and/or configuration properties if required for development.	Tier 2	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-041	A DK shall document all licensed/non-redistributable tools and libraries required for the development environment.	Tier 1	Inspection	Yes	Yes	Yes	Yes	
GRA-DK-042	A DK shall document all licensed/non-redistributable software required for the build environment.	Tier 1	Inspection	Yes	Yes	Yes	Yes	Applies only to DPP
GRA-DK-043	A DK shall document all licensed/non-redistributable software required for the build environment.	Tier 3	Inspection	Yes	Yes	Yes	Yes	Applies only to MFA
GRA-DK-044	A DK shall identify specific versions of commodity packaging scripts and/or tools.	Tier 3	Inspection	Yes	Yes	Yes	Yes	Applies only to MFA

Table 5.3.1: A DK's Adherence to the Compliance Requirements Determines Its Compliance Score.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
API	Application Programming Interface
ASB	Abstract Service Bus
BIT	Built-In Test
CCF	Compute Cluster Farm
CDS	Cross Domain Solution
CI/CD	Continuous Integration/Continuous Delivery
CMM	Countermeasures Manager
COARPS	Common Open Architecture Radar Program Specification
CofC	Consequence of Compromise
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
DaR	Data at Rest
DAST	Dynamic Application Security Testing
DevSecOps	Development, Security, and Operations
DGT	Distributed Guaranteed Time
DK	Development Kit
DP	Digital Payload
DPP	Digital Payload Processor
FACE	Future Airborne Capability Environment
FPGA	Field-Programmable Gate Array
GOTS	Government-Off-The-Shelf
GPP	General Purpose Processor
GPU	Graphics Processing Unit
GRA	Government Reference Architecture
IaC	Infrastructure as Code

IAW	In Accordance With
ICD	Interface Control Document
IDD	Interface Description Document
IEEE	The Institute of Electrical and Electronics Engineers
KMP	Key Management Plan
MASI	Mission Agnostic Service Infrastructure
MBSE	Model-Based Systems Engineering
MEL	MFA Encapsulation Layer
MFA	Multi-Function Aperture
MFP	Multi-Function Processor
MIL-STD	Military Standard
MLS	Multi-Level Security
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NFS	Network File System
NSA	National Security Agency
OCI	Open Container Initiative
OEM	Original Equipment Manufacturer
OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
PTP	Precision Time Protocol
RAM	Random Access Memory
RoT	Root of Trust
SAST	Static Application Security Testing
SBOM	Software Bill of Materials
SK	Starter Kit
SM	Security Module
SMS	Stores Management System
UCI	Universal Command and Control Interface

VA	Virtual Aperture
VITA	VME bus International Trade Association
VME	Versa Module Eurocard
WSI	Weapon Systems Interface
YAML	YAML Ain't Markup Language

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision adds prose updates and clarifications.	July 24, 2025	SK13.0
B	This revision adds prose updates and updates to the MPU and MS requirements.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Compliance Assessment Approach Volume Between Documentation Releases.